1) **List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep)**

Hardware Issue	Typical Symptom
1. No display on monitor	Blank screen, no POST beep, monitor shows "No Signal."
2. Faulty RAM	PC beeps repeatedly during startup, random crashes, or fails to boot.
3. Hard drive/SSD failure	Operating system does not load, "Boot Device Not Found" error, unusual clicking noises (HDD).
4. Overheating CPU	PC shuts down unexpectedly, freezes, restarts, or runs very slowly with loud fan noise.
5. Power supply (PSU) failure	PC does not turn on, no lights or fan activity, or powers off suddenly during use.

2) Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Step-by-step troubleshooting plan

1. Check the power outlet
   - Plug another working device into the same outlet.
   - Check: Confirm the wall socket is supplying power.
2. Inspect the charger/adapter
   - Check the charger cable, adapter, and connectors for cuts, burns, loose wires, or physical damage.
   - Check: Make sure the adapter is not damaged and the charging connector is firmly attached.
3. Check the charging indicator
   - Connect the charger to the laptop and look for the charging LED.
   - Check: If there is no LED, the problem could be the charger, charging port, battery, or motherboard.
4. Check the charging port
   - Carefully inspect the laptop's DC charging jack for dust, bent pins, looseness, or damage.
   - Check: Gently move the connector (without forcing it) to see whether the charging LED flickers.
5. Test with another compatible charger
   - If available, use a known-good charger with the same voltage, connector type, and suitable current rating.
   - Check: If the laptop charges, the original charger is likely faulty.
6. Remove and reconnect the battery (if removable)
   - Turn off the laptop, disconnect the charger, remove the battery, and reinstall it.
   - Check: Look for swelling, leakage, or physical damage. Do not use a swollen battery.

3) Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

Step	Procedure	Observation
1	Set the multimeter to **DC Voltage (V⎓)**.	Meter is ready to measure DC voltage.
2	Connect the charger to a power outlet.	Charger is powered.
3	Place the **black probe on the USB ground (GND)** contact and the **red probe on the +5V (VBUS)** contact.	Multimeter displays the output voltage.
4	Record the reading.	**Measured voltage: 5.02 V DC**
5	Compare the reading with the expected USB range.	5.02 V is between **4.75 V and 5.25 V**.

**4) A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.**

Multimeter Test for ATX 12V Rail

1. Shut down the PC and unplug the power cable from the wall.
2. Disconnect the ATX power connector from the motherboard.
3. Set the multimeter to DC Voltage (V⎓).
4. Turn the PSU on only if you are using an appropriate safe PSU-testing setup. Do not probe an energized connector casually.
5. Identify the yellow wire (+12V) and black wire (GND) on the ATX connector.
6. Place the red probe on the yellow wire/pin and the black probe on a black wire/pin.
7. Record the displayed voltage.